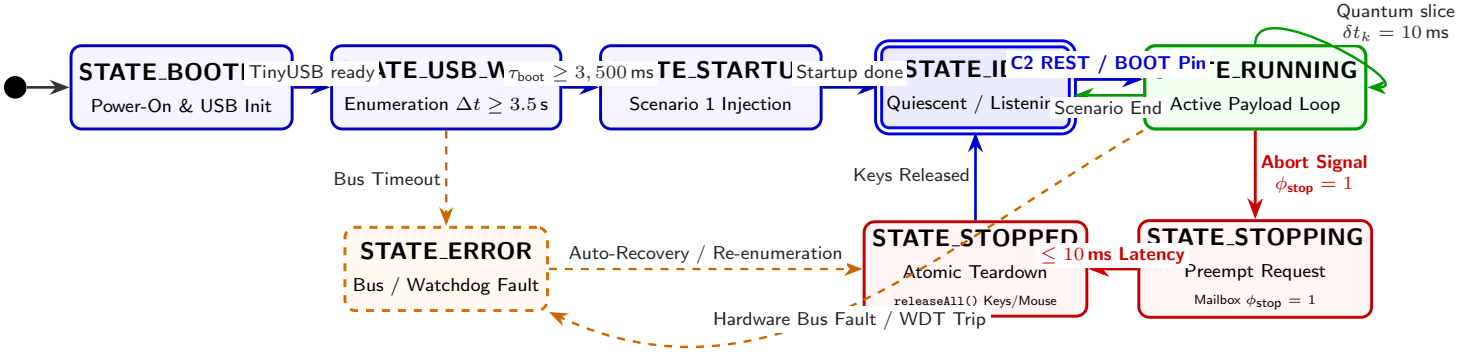

Quantum slice
$\delta t_k = 10\,\text{ms}$
STATE_BOOT
Power-On & USB Init
TinyUSB ready
ATE_USB_W
Enumeration $\Delta t \geq 3.5\,\text{s}$
$\tau_{\text{boot}} \geq 3,500\,\text{ms}$
TE_STARTU
Scenario 1 Injection
Startup done
STATE_II
Quiescent / Listenir
C2 REST / BOOT Pin
E_RUNNING
Scenario End
Active Payload Loop
Abort Signal
$\phi_{\text{stop}} = 1$
Keys Released
Bus Timeout
STATE_ERROR
Bus / Watchdog Fault
Auto-Recovery / Re-enumeration
STATE_STOPPED
Atomic Teardown
releaseAll() Keys/Mouse
≤ 10 ms Latency
STATE_STOPPING
Preempt Request
Mailbox $\phi_{\text{stop}} = 1$
Hardware Bus Fault / WDT Trip